

Table 12.13 (Output 5) has two parts of analysis: reproduced correlation and residuals. The reproduced correlation is the correlation among the extracted components. From this table, we intend to ensure that the correlation between the original variables and reproduced matrix should be as close as possible. The lower part of the table 'residual matrix' shows the difference between original variable and residual matrix. For a good PCA, it was expected that the difference between original variable and extracted matrix should be near to zero. Once the difference is near to zero, it can be said that the extracted components accounted a larger variation in the original correlation matrix. Here we can take an example from the table that the original correlation between irrigation and other than agricultural labour is 0.403, Whereas the extracted correlation between these two variable is 0.554 and the difference between the two correlation given in residual part is $0.403 - 0.554 = -0.149$.

Final PCA Index Value

The final index is calculated by the addition of multiplication of normalized value of the variable and the Eigen vector of that variable (first component). In our example the PCA index value for all the 11 selected district can be calculated by using two Table 12.5 and Table 12.12 given above. The final PCA index of first district (Malkangiri) is

PCA index Malkangiri District = $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n$

$$= (0.202 \times 1.237) + (-0.185 \times 1.332) + (-0.403 \times 1.899) + (0.717 \times 0.808) + (0.837 \times 0.561) + (-0.721 \times 1.079) + (0.504 \times 1.379) + (0.733 \times 0.614) + (0.441 \times 0.775) + (-0.430 \times 1.003) = 0.049$$

Likewise the PCA Index value of all other districts can be calculated as given in Table 12.14

Table 12.14: PCA Index Value

District	PCA Index
Malkangiri	0.049
Rayagada	1.273
Sundargarh	1.322
Nabarangapur	1.145
Kandhamal	1.641
Koraput	0.517
Mayurbhanj	1.333
Gajapati	1.757
Sambalpur	0.465
Kendujhar	0.290
Jharsuguda	1.551

The table 12.14 reveals that Gajapati is the most developed district followed by Kandhamal district. On the other hand, Malkangiri is most backward district.

12.6.2 Use of Output Indicators

After finding out the Index findings need to be validated. This can be done by comparing the result with some output indicators. Let us illustrate with example discussed in section 12.6. District wise final index as arrived out by PCA method is given in Table 12.14. This finding can be validated by taking an output indicator say infant mortality rate (IMR). We can run a correlation between PCA index and the output indicator 'Infant mortality rate'. This has been provided in Table 12.15 which is 0.893. The table shows that the most backward district has a higher degree of mortality. Hence our final index is validated. If suppose the correlation between these two variable is -0.120, then our index is not validated as the correlation between the two variable is very low.

Table 12.15: PCA Index and Comparing with Output Indicators

District	PCA Index	Infant Morality Rate*
Malkangiri	0.049	120
Kendujhar	0.290	100
Sambalpur	0.465	110
Koraput	0.517	92
Nabarangapur	1.145	89
Rayagada	1.273	100
Sundargarh	1.322	80
Mayurbhanj	1.333	75
Jharsuguda	1.551	65
Kandhamal	1.641	60
Gajapati	1.757	55

Note: Imaginary numbers

12.6.3 Use of Weight

Assigning weight i.e. all the indicators/dimensions are treated equally or differentially is an important issue in construction of composite index.. Sometimes different dimensions are given differential weight but in construction of overall index, an equal weight approach is followed. The human development index assigns differential weights to indicators/dimension but for overall index, an equal weight approach is followed. In construction of human development index, in 2001 by Planning Commission, the overall index was calculated by using equal weight whereas the different sub-indices like Composite indicator on educational attainment, Composite indicator on health attainment differential weights were given. In calculating educational attainment, two variables i.e. 'literacy rate for the age group 7 years and 'adjusted intensity of formal education' were used where the first and second variable were given 1/3 and 2/3 weight respectively. The other 'health attainment was calculated by taking two variables 'life expectancy at age one,'

and 'infant mortality rate'. The first variable has a weight of 2/3 whereas the second variable 1/3. But the overall weight is calculated by taking equal weight for both the variables. Take another example of our range equalization method where we have given equal weight to all variables. In some other cases, the individual variables of sub indices were given equal weights but the overall index was calculated by giving differential weights to sub-indicators.

Let us denote the weight in mathematical notation:

$$I=XW,$$

where 'X' is a matrix with m rows and n column, 'I' is identity matrix and 'W' is weight. Hence the final indices can be arrived at by taking the weighted component/variables. There are basically two ways in assigning weights. According to Munda and Nardo (2005), we can define weight by taking the importance of the particular variable or group of variables. In this process, the weights are arrived at by the past knowledge or observation of individual and the probable effects of that variable in the overall analysis. Let us illustrate with an example. Suppose we want to find out the sanitation and hygiene Index. For this we have selected some variables like 'proportion of people having access to toilet', 'proportion of people having drainage', 'proportion of people with wash hand', 'proportion of people safely disposed child exgratia' etc . Here either on the basis of literature or on the basis of our observation from some villages, we found that not having toilet is very important then wash hand before eating. In this case we can put higher weight for having toilet and less weight for washing hand.

In case of literature, weights are derived from on the basis of theoretical or statistical consideration.

12.6.4 Limitations of Principal Component Analysis

The major criticisms against the use of PCA is that the technique is arbitrary in constructing indices. The number of components and the number of variables used are not well defined. This method entirely depends on the first component. If the first component does not explain large variation, this method is not useful. Such possibility depends on nature of data and on relationship of variable.

Alternative methods to PCA that can reduce the dimensionality of the data include: correspondence analysis, multivariate regression or factor analysis. The details about these methods have been provided in Unit 13 and Unit 16 of Block 4.

12.7 MERITS AND LIMITATIONS OF COMPOSITE INDEX

One of the important merits of a composite index is that this can summarize the complex and multidimensional indicators into one indicator which helps the policy makers for implementation of a particular programme or policy decision. A composite index can easily be taken to interpret about the development or backwardness of a particular region or area or sector. The progress of a state in India say, e.g. Bihar can easily be accessed and we can compare the status of Bihar at two or more points of time. The composite index

can reduce the number of indicators without reducing the underlying information base. The composite index facilitates communication to general public and promote accountability. The composite index is not free from limitations. It has the risk of misleading policy message if it is poorly constructed or misinterpreted. The calculation may be misleading and faulty if sound statistical or conceptual principle is not applied. The assignment of weight many times creates a debate. One of the principles of assigning weight is on the basis of value judgment. This always being a source of criticism. Again allocation of upper and lower bound (mainly in range equalization method) is a point of criticism.

Check Your Progress 3

- 1) In the context of Composite Index, what are the uses of Principal Component Analysis (PCA).

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- 2) What is explained by the term 'communality'?

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- 3) How is Final Index Value calculated?

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- 4) In construction of Human Development Index, how weights are assigned to the different indicators?

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5) State the limitations of Composite Index.

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12.8 LET US SUM UP

Composite index is an important analytical technique to analyse the developmental related issues like status of development, food security, human development etc. at the district/region/state level. It is an expression of single score made of different scores measuring the various dimensions and indicators of particular issue. This Unit describes the process to derive composite index by applying the various methods. These methods include simple ranking method, indices method, mean standardisation method, and range equalization method. The main advantage of the rank and indices method is that they are easy to understand and interpret. The RE, MS and PCA methods uses all the variables in reducing the dimensionality of the data. These methods are very much useful in comparing across districts, states, countries or areas such as rural and urban. Again these methods are useful to compare over time by constructing composite index at two points of time by taking same indicators/variables. The principal component analysis enables the students to reduce the indicators that are uncorrelated to explain the variation in the original data set. To run the PCA, SPSS and STATA softwares are preferred.

12.9 EXERCISES

- 1) Think and collect some gender related variables from the data sources like women literacy, women workforce participation rate in the age group 15-59, mean year schooling, mortality rate, immunization rate etc and find out the gender development index.
- 2) Explain the process of using Principal Component Analysis (PCA) to reduce the number of variables in a data set to smaller number of dimensions.
- 3) By using range equalization method find out the food security index from the report prepared by IHD-WFP given in website
'<http://122.180.7.122/displaymorePub.asp?itemid=84&subchkey=11&chna me=Publications>.'

12.10 SOME USEFUL BOOKS/REFERENCES

A. Saltelli, G. Munda and M. Nardo (2006), 'From Complexity to Multidimensionality: the Role of Composite Indicators for Advocacy of EU Reform', Tijdschrift voor Economie en Management Vol. LI, 3

Planning Commission (2001), 'National Human Development Report', Government of India downloaded from <http://planningcommission.nic.in/reports/genrep/index.php?repts=nhdcont.htm>

OECD (2008), 'Handbook on Constructing Composite Indicators Methodology and User Guide', Organization For Economic Co-operation and Development.

12.11 ANSWER OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See Section 12.2
- 2) See Section 12.2
- 3) See Section 12.3
- 4) See Section 12.4

Check Your Progress 2

- 1) See Sub-section 12.4.1
- 2) See Sub-section 12.4.2
- 3) See Sub-section 12.5.4

Check Your Progress 3

- 1) See the heading 'conducting PCA and analyzing result' under Section 12.6
- 2) See Sub-section 12.6.1 (Under the head conducting the PCA and analyzing results)
- 3) See Sub-section 12.6.1 (Under the head final PCA index value)
- 4) See Sub-section 12.6.3
- 5) See Section 12.7